

Fig. 2: Flow chart

ASCII format for display. Only the commands are sent in hex form to the LCD. Register-select RS signal is used to distinguish between data (RS=1)

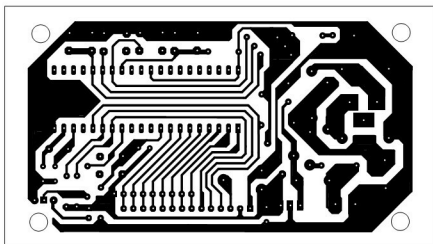


Fig. 3: An actual-size, single-side PCB for the microcontroller-based digital alarm clock

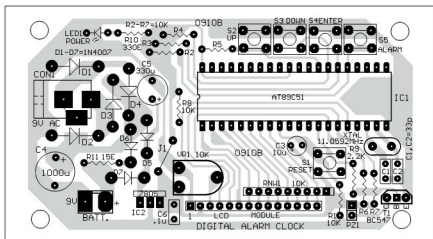


Fig. 4: Component layout for the PCB

and command (RS=0). Using preset VR1 you can control the contrast of the LCD.

Switch on the power to the system and enter the current time with the help of switches S2 through S4. 'Up,' 'down' and 'enter' switches are connected to pins P2.7, P2.6 and P2.5 of microcontroller AT89C51, respectively. Using 'up' switch, you can increase the numeral values of hours, minutes and seconds, respectively, for time or alarm setting, while 'Down' switch can be used to decrease the values. The current time or alarm time set using up/down switches is accepted on pressing 'enter' switch. The time is set and displayed in 12-hour (a.m./p.m. format).

Using 'alarm' switch connected to pin P2.4 of the microcontroller you can

activate or deactivate the alarm function. When the alarm is activated, enter the new alarm time in the same way as the current time. The message "Alarm Active" is displayed on the second line of the LCD with current running time on first line of the LCD. Every second, current time is compared with the alarm time. When current time matches with alarm time, pin P3.3 of the microcontroller goes high to drive the transistor into saturation. Piezobuzzer PZ1 sounds until you press the alarm switch (S5) to deactivate the alarm.

The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver secondary output of 12V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C4